Angled launches

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use components to analyze an angled projectile.

SUCCESS CRITERIA - I can...

1. Separate the initial velocity into two components.
2. Use vertical motion to determine flight time.
3. Explain what must be true for your range comparisons to work.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A downward-velocity fit goes from **0.3 m/s** at **0.01 s** to **3.24 m/s** at **0.31 s**. **Find g** and **explain why fitting an interval** can be better than one noisy point.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $v_{x,\,i} = v_i \cos\theta$; $v_{y,\,i} = v_i \sin\theta$; $t_{flight} = 2\, v_{y,\,i}/g$
range = $v_{x,\,i}\, t_{flight}$.
θ above horizontal; g = **10 m/s²**. Equal launch/landing height; no air resistance.

Worked reasoning: A projectile has initial components **4 m/s** horizontal and **10 m/s** upward. With g=**10 m/s²** it reaches the top in **1 s** and returns to launch height in **2 s**; range=**8 m**.

Your turn: Sketch its path and velocity arrows at launch, top and return.



Add velocity / acceleration arrows.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

At the highest point, **which velocity** component is zero and which remains? Is acceleration zero?



Add velocity / acceleration arrows.

Marzano check - circle: ① ② ③

04 Apply the physics

A ball launches from level ground at **20 m/s**, **30°** above horizontal, and lands at the same height.
1. Use g = **10 m/s²** and ignore air resistance.
2. **Draw the initial velocity components**.
3. **Find flight time**, range and maximum height.



Add velocity / acceleration arrows.

Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

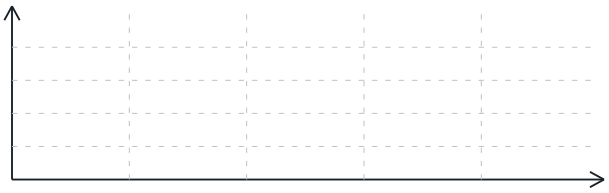
In this ideal projectile calculation, why can you double the time to the top only when the object lands at its launch height?

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

A simplified calculation predicts position (**-70**, **90**) km with bounds ±**7 km** in x and ±**9 km** in y. **Write a short briefing giving** both intervals and two reasons it is not a real asteroid collision forecast.
Label the given and predicted quantities on the diagram.

y position (km)



x position (km)

Marzano check - circle: ① ② ③

07 Exit check

For speed **20 m/s**, compare ranges at **30°** and **60°** on level ground. **Explain which has longer flight** time. **Label the diagram** with the known quantities before calculating. Ignore air resistance; use g = **10 m/s²** when needed.

Separate horizontal and vertical motion

Label x and y; state positive directions.

Velocity:

Acceleration / force:

Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

For a launch at **20 m/s** from level ground, find the two angles giving an ideal range of **20 m** with g=**10 m/s²**. **Compare flight times**. **Label the diagram** with the known quantities before calculating. Ignore air resistance; use g = **10 m/s²** when needed.



Add velocity / acceleration arrows.

Marzano check - circle: ① ② ③